

Design Proposal

William Huynh

edX public username: HuynhWSiPho

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Abstract

This proposal describes four passive silicon Mach–Zehnder interferometers (MZIs) contained in `EBeam_HuynhWSiPho.gds`. The objective is to extract the quasi-TE waveguide group index from the free spectral range (FSR) and compare it with a simulated waveguide model. Each circuit uses nominally 500 nm-wide strip waveguides, a Y-branch input splitter, a broadband directional-coupler combiner, and three TE grating couplers. The common short arm is $31.282\ \mu\text{m}$; the four arm imbalances are 68.717 , 118.719 , 168.717 , and $58.717\ \mu\text{m}$. With the course-specified 220 nm silicon thickness and an approximate oxide-clad effective-index model, $n_g(1550\text{ nm}) \approx 3.8935$. The corresponding FSRs range from 3.66 to 10.51 nm, providing several fringes within a 50 nm scan. A compact polynomial model, normalized circuit spectra, and a numerical check of the group-index extraction are presented. The simulations are estimates and will have to be compared to experimental data.

1 Introduction and design objectives

The group index sets the delay per unit length of a guided optical signal and is a useful quantity for designing delay lines, filters, interferometers, and resonators. Whereas the effective index determines accumulated phase, the group index describes how that phase changes with optical frequency. An unbalanced interferometer converts this frequency-dependent phase into measurable transmission fringes. Its FSR therefore provides a practical way to characterize the waveguide without directly measuring propagation time.

The objectives are to (i) measure the quasi-TE group index of a nominal $500\text{ nm} \times 220\text{ nm}$ silicon strip waveguide, (ii) verify the inverse dependence of FSR on arm imbalance, and (iii) compare the extracted group index and its wavelength dependence with waveguide simulations. The selected devices vary path length while keeping nominal width, polarization, and component types fixed. This separates the path-length trend from an intentional width sweep.

The supplied file contains four designs. If more room was provided, an updated layout would have a grating-to-grating reference waveguide which would aid in repeatability and normalization. These are proposed additions and are not present in the uploaded GDS.

2 Layout-derived circuit design

2.1 Evidence and assumptions

The physical top cell is `EBeam_HuynhWSiPho.v1`. The layout contains MZI1–MZI4 in a $605\ \mu\text{m} \times 410\ \mu\text{m}$ floorplan. Its silicon geometry and routing-marker overlays are shown in Fig. 1.

The waveguide itself is 500 nm in width, and has a nominal bend-radius parameter of $5\ \mu\text{m}$. The silicon thickness is $h = 220\text{ nm}$ and is an industry standard. An oxide upper and lower cladding is assumed for the numerical model, consistent with the oxide grating-coupler cell name. Actual thicknesses, oxide coverage, and material dispersion must be confirmed after fabrication. For example, some of these quantities can be measured using a SEM and measurements of the device (both of and on) will be compared towards the end of the report.

2.2 Components and optical connections

Each MZI contains one `ebeam_y_1550` Y-branch, one `ebeam_bdc_te1550` broadband directional coupler, and three `ebeam_gc_te1550` TE grating couplers. The input grating feeds the Y-branch. Its two outputs connect through unequal waveguide arms to the two input ports of the directional coupler, which is rotated by 90° . Two further waveguides connect the directional-coupler outputs to the output gratings. The circuit is therefore a Y-branch/broadband-coupler MZI.

The directional-coupler label identifies a nominal 50/50 TE splitter at 1550 nm and cites the asymmetric-waveguide phase-control design of Lu *et al.* [2]. The grating cell references Wang *et al.* [3]; these references identify the component design lineage rather than establish the loss or bandwidth of the specific fabricated devices. No measured component S-parameters were supplied.

For every device, the local input grating port is at $(39.969, 267.500)\ \mu\text{m}$. The output grating ports are at $(39.969, 140.500)\ \mu\text{m}$ and $(39.969, 13.500)\ \mu\text{m}$, giving $127\ \mu\text{m}$ pitch. Figure 2 shows two representative circuits. Layer-10 input labels have the form `opt_in_TE_1550_device_HuynhWSiPho_MZI1`, with the device number changed for each cell.

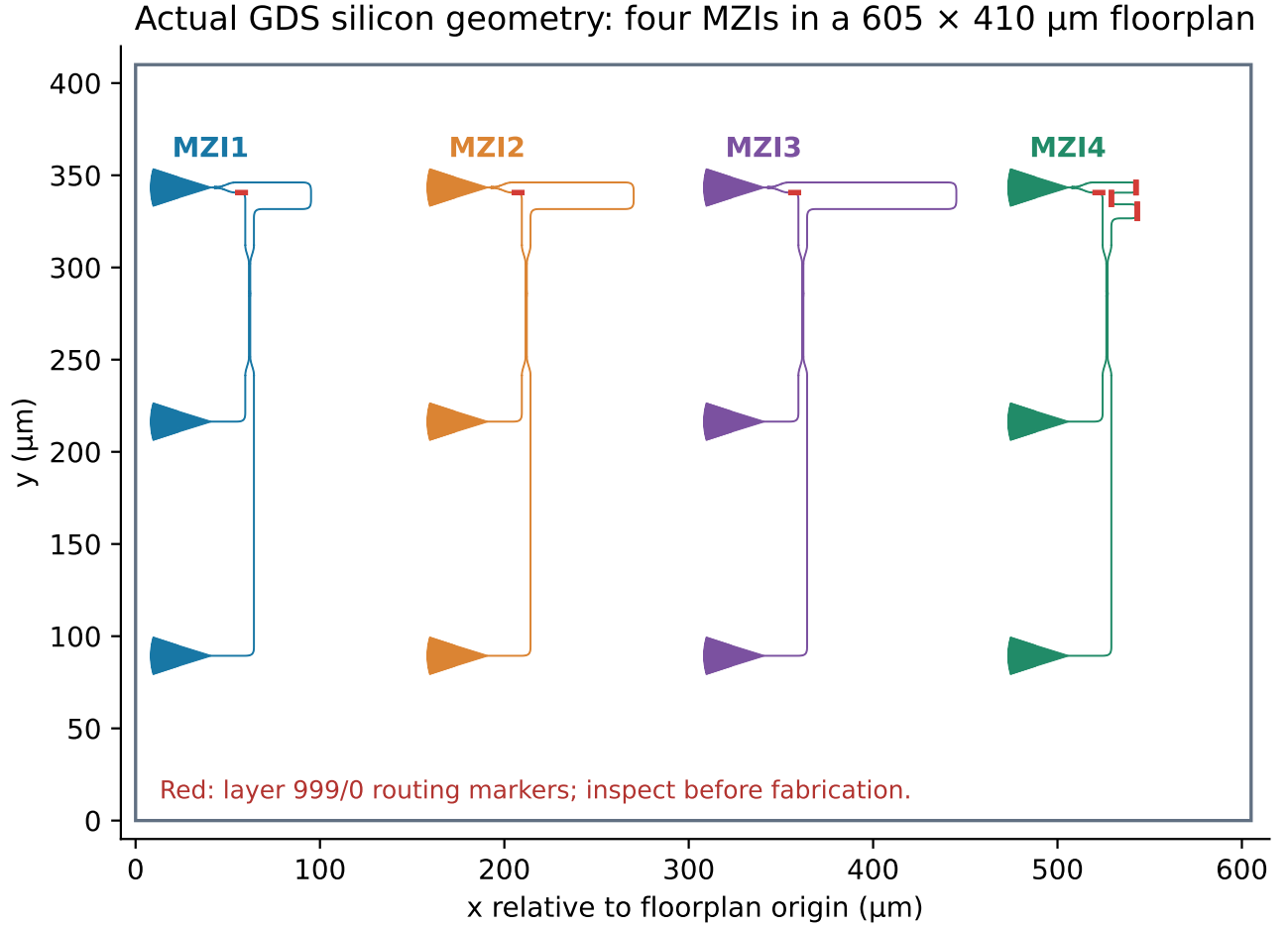


Figure 1: Cartoon images of the device layout. Devices appear from left to right as MZI1, MZI2, MZI3, and MZI4. The frame is the actual $605 \mu\text{m} \times 410 \mu\text{m}$ floorplan. Red overlays indicate paths on layer 999/0, which coincide with short routing segments and require review.

2.3 Arm lengths and parameter variations

The relevant lengths are the two routes between the Y-branch output ports and the directional-coupler input ports. They were read from the embedded `Spice_param:wg.length` labels and corroborated by the `Length` labels, rather than estimated from bounding boxes or sharp-corner waypoint distances. They are PCell-reported route lengths, conditional on the generated geometry being accepted; they are not independently measured fabricated lengths. This would have to be compared to if we were to measure device dimensions post-fabrication!

The common short-arm cell Waveguide is $31.282 \mu\text{m}$ long. Long-arm cells are Waveguide\$4, Waveguide\$5, Waveguide\$6, and Waveguide\$8 for MZI1–MZI4. The output access routes are 42.378 and $174.078 \mu\text{m}$ long. Their difference is not included in ΔL : they occur after recombination and cannot create the principal MZI arm interference. They can affect output attenuation and, if reflections are appreciable, parasitic interference.

The shortest imbalance, MZI4, provides the largest FSR and the fewest fringes. MZI3 provides the smallest FSR and the

Table 1: Actual arm-length variation and approximate performance at 1550 nm ; $n_g = 3.893484$. All devices use $w = 500 \text{ nm}$, $h = 220 \text{ nm}$, quasi-TE polarization.

Device	L_2 (μm)	ΔL (μm)	FSR (nm)	50nm/FSR
MZI1	99.999	68.717	8.980	5.57
MZI2	150.001	118.719	5.198	9.62
MZI3	199.999	168.717	3.657	13.67
MZI4	89.999	58.717	10.509	4.76

most fringe intervals for extracting dispersion. All four satisfy $\text{FSR} < 50 \text{ nm}$ within this model. The minimum imbalance for one 50 nm FSR near 1550 nm is approximately

$$\Delta L_{\min} \approx \frac{(1.55 \mu\text{m})^2}{3.893484(0.050 \mu\text{m})} = 12.34 \mu\text{m}. \quad (1)$$

The existing designs exceed this minimum by factors of about 4.8–13.7. The ratios in Table 1 count approximate fringe periods across the band, not a guaranteed integer count of peaks at arbitrary scan endpoints.

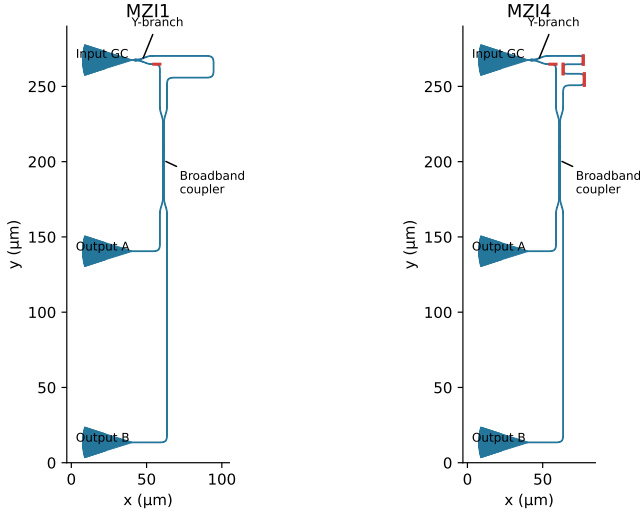


Figure 2: Actual GDS geometry of MZI1 and MZI4. Input light enters the upper grating. The Y-branch splits the light into two arms; the broadband coupler recombines it into two outputs. Red routing markers are retained for review.

3 Theory

3.1 Propagation phase and group index

For a waveguide mode with propagation constant β ,

$$\beta(\lambda) = \frac{2\pi n_{\text{eff}}(\lambda)}{\lambda}, \quad n_g = n_{\text{eff}} - \lambda \frac{dn_{\text{eff}}}{d\lambda}. \quad (2)$$

The second relation follows from $n_g = c d\beta/d\omega$ with $\omega = 2\pi c/\lambda$. For equal waveguide cross-sections in the two arms, the differential phase is

$$\Phi(\lambda) = \beta(\lambda)\Delta L + \phi_c(\lambda), \quad (3)$$

where ϕ_c includes the relative phase introduced by the splitter and combiner. A wavelength-independent ϕ_c shifts fringe positions but does not change the local FSR.

3.2 Transfer function

Let a_1 and a_2 be the positive arm-to-output field-amplitude factors, including input splitting, output combining, and propagation attenuation. A general output transmission is

$$T(\lambda) = a_1^2 + a_2^2 + 2a_1a_2 \cos \Phi(\lambda). \quad (4)$$

For ideal equal splitting and combining, power attenuation coefficient α in reciprocal length units gives $a_j = \frac{1}{2}e^{-\alpha L_j/2}$, hence

$$T_A = \frac{1}{4} [e^{-\alpha L_1} + e^{-\alpha L_2} + 2e^{-\alpha(L_1+L_2)/2} \cos \Phi]. \quad (5)$$

In the lossless case, the two complementary outputs are

$$T_A = \frac{1 + \cos \Phi}{2}, \quad T_B = \frac{1 - \cos \Phi}{2}. \quad (6)$$

These expressions absorb the actual coupler phase into ϕ_c ; a mixed Y-branch/directional-coupler circuit can have a different phase origin from a two-identical-coupler MZI. The calculation below sets $\phi_c = 0$ to illustrate the spectra and test FSR extraction. It does not predict the absolute peak wavelengths of the real circuit.

The fiber-to-fiber transmission is further multiplied by the input and output grating-coupler efficiencies and output-route attenuation. Unequal arm amplitudes reduce fringe visibility,

$$V = \frac{T_{\text{max}} - T_{\text{min}}}{T_{\text{max}} + T_{\text{min}}} = \frac{2a_1a_2}{a_1^2 + a_2^2}. \quad (7)$$

Loss and visibility cannot be reliably forecast from the mask alone, so the presented spectra are normalized lossless predictions, not insertion-loss predictions.

3.3 Derivation of group index from FSR

For adjacent maxima, $|\Delta\Phi| = 2\pi$. Differentiating Eq. 3 and initially treating ϕ_c as constant yields

$$\frac{d\Phi}{d\lambda} = -\frac{2\pi n_g \Delta L}{\lambda^2}. \quad (8)$$

Across a sufficiently narrow fringe interval,

$$\text{FSR}_\lambda \approx \frac{\lambda^2}{n_g |\Delta L|}, \quad \boxed{n_g \approx \frac{\lambda^2}{|\Delta L| \text{FSR}_\lambda}}. \quad (9)$$

All lengths in Eq. 9 must use the same units. In frequency, $d\Phi/df = 2\pi n_g \Delta L/c$, giving

$$\text{FSR}_f \approx \frac{c}{n_g |\Delta L|}. \quad (10)$$

This is the standard unequal-arm MZI relation used in circuit modeling [1].

For adjacent peak wavelengths $\lambda_a < \lambda_b$, use the measured frequency separation instead of approximating it by a center-wavelength conversion:

$$\Delta f = c \left(\frac{1}{\lambda_a} - \frac{1}{\lambda_b} \right), \quad (11)$$

$$\boxed{\bar{n}_g = \frac{\lambda_a \lambda_b}{|\Delta L| (\lambda_b - \lambda_a)}}. \quad (12)$$

With constant coupler phase, Eq. 12 exactly gives the group index averaged over that frequency interval, even if n_g varies across it. It is approximately the local group index at the interval midpoint when dispersion is weak. A peak-to-trough spacing is half an FSR and must not be substituted for a peak-to-peak spacing.

If the combiner phase is dispersive, the extracted quantity is instead an apparent index,

$$n_g^{\text{app}} = n_g + \frac{c}{2\pi \Delta L} \frac{d\phi_c}{df}. \quad (13)$$

More generally the phase slope contains the differential sum of all arm-section group delays, including bends and unequal-width sections. A straight-waveguide index is obtained only when those extra differential contributions cancel or are modeled.

4 Modeling and numerical simulations

4.1 Approximate waveguide model

A two-step effective-index method (EIM) is used as an explicitly approximate calculation. First, the fundamental TE mode of a symmetric 220 nm silicon slab is solved vertically. Its effective index becomes the core index of a lateral 500 nm slab, for which the fundamental TM slab relation is used. This TE-vertical/TM-lateral choice approximates a quasi-TE rectangular guide with the dominant electric field along the width. Both slab exteriors use oxide index 1.444. Silicon index is fixed at 3.476 across the scan. These values are chosen model inputs; material dispersion, sidewall slope, finite oxide thickness, substrate leakage, and corner-vector effects are omitted.

For slab thickness t , indices n_c, n_s , half-thickness $a = t/2$, and fundamental even mode,

$$V_s = \frac{\pi t}{\lambda} \sqrt{n_c^2 - n_s^2}, \quad (14)$$

$$u \tan u = q \sqrt{V_s^2 - u^2}, \quad 0 < u < \min(V_s, \pi/2), \quad (15)$$

$$n_{\text{slab}}^2 = n_c^2 - \left(\frac{u\lambda}{\pi t} \right)^2, \quad (16)$$

where $q = 1$ for TE and $q = n_c^2/n_s^2$ for TM. Root finding on the fundamental branch avoids substituting a higher-order slab mode. The wavelength sweep covers 1525–1575 nm with 501 points; $dn_{\text{eff}}/d\lambda$ is evaluated with a centered $10^{-5} \mu\text{m}$ step. The solver root tolerance is 10^{-14} in u .

At 1550 nm this model yields

$$n_{\text{eff}} = 2.491433, \quad n_g = 3.893484. \quad (17)$$

These are estimates within the chosen EIM and fixed-material assumptions, not full-vector quasi-TE predictions. In particular, the small computed change in n_g across the band must not be interpreted as a validated low-dispersion result. The compact-fit error and numerical extraction error below test internal numerical consistency, not EIM accuracy.

For the profile in Fig. 3, each slab has a cosine core envelope and an evanescent tail,

$$F(s) = \begin{cases} \cos(us/a), & |s| \leq a, \\ \cos u e^{-v(|s|/a-1)}, & |s| > a, \end{cases} \quad (18)$$

with $v = \sqrt{V_s^2 - u^2}$. The plotted separable approximation is $\psi(x, y) = F_x(x)F_y(y)$. This visualizes confinement, but does not reproduce all vector-field boundary conditions or establish that no higher-order rectangular-guide modes exist.

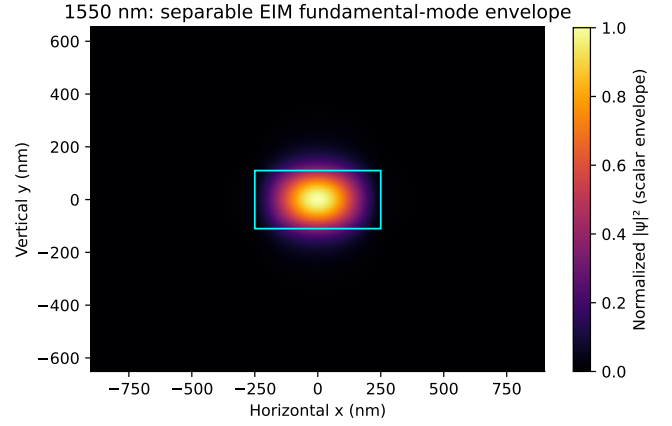


Figure 3: Approximate fundamental-mode scalar envelope at 1550 nm. Cyan outlines the 500 nm $\times$ 220 nm silicon core. The plotted quantity is $|\psi|^2$ normalized to its maximum, not a full-vector electric-field intensity or a Lumerical MODE export.

4.2 Effective/group index and compact model

Figure 4 presents the computed effective and group indices. Define $p = \lambda_{\text{um}} - 1.55$, where λ_{um} is the numerical wavelength in micrometers. A least-squares cubic fit gives

$$n_{\text{eff}}(p) = 2.4914332285 - 0.9045490711p + 0.0012317028p^2 + 0.1498484747p^3. \quad (19)$$

Equivalently, if dimensional p is used, the nonconstant coefficients have units μm^{-1} , μm^{-2} , and μm^{-3} . This fit is intended only for 1.525–1.575 μm and the specified cross-section. The largest effective-index residual over the 501 sample points is 5.34×10^{-9} relative to the EIM data. The fitted group index is obtained analytically from

$$n_g(p) = n_{\text{eff}}(p) - (1.55 + p)(a_1 + 2a_2p + 3a_3p^2), \quad (20)$$

where a_1, a_2, a_3 are the corresponding polynomial coefficients in Eq. 19. No independent group-index fit is required.

4.3 Circuit transmission spectra

The compact model is substituted into Eqs. 3 and 6, using the actual imbalances from Table 1. Spectra are sampled at 20,001 wavelengths across the 50 nm band. Figure 5 displays output A; output B is complementary under the lossless assumptions. The fringe density increases as ΔL increases, while the same waveguide dispersion appears in all devices. The normalized ideal visibility is unity by assumption, not an experimental extinction-ratio prediction.

4.4 Numerical validation of the extraction

Adjacent maxima near 1550 nm were found by solving $\Phi(\lambda) = 2\pi m$ with the directly evaluated EIM index, independently of the sampled spectral grid. Equation 12 was then applied to the peak pairs. Table 2 compares the inferred index with the EIM

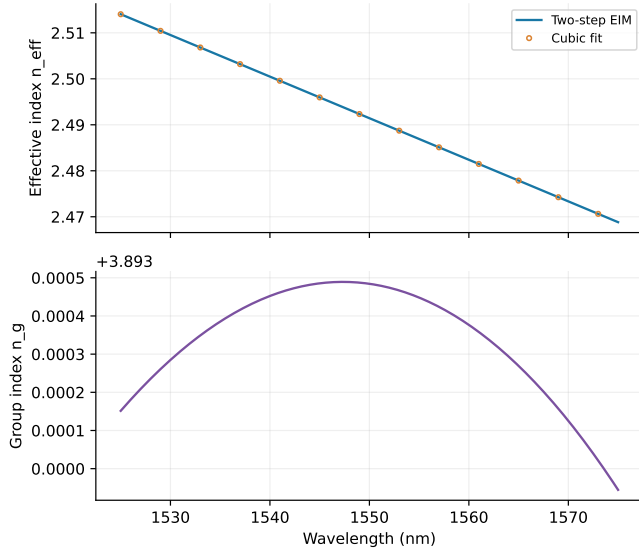


Figure 4: Effective and group index versus wavelength for the approximate oxide-clad EIM model. Open markers compare the cubic fit with the directly solved effective index. Material indices are held fixed; the plotted dispersion is only the model's geometrical waveguide dispersion.

Table 2: Group-index extraction from simulated adjacent maxima. Peak wavelengths are in nm; values shown are rounded from the calculation.

	λ_a	λ_b	Extracted n_g	Midpoint n_g
MZI1	1545.113	1554.088	3.893481	3.893486
MZI2	1549.097	1554.306	3.893474	3.893476
MZI3	1547.049	1550.701	3.893487	3.893488
MZI4	1543.515	1554.007	3.893481	3.893488

derivative at each wavelength midpoint. The largest relative difference is below 0.00017%, consistent with interval averaging and weak model dispersion. This demonstrates that the extraction equation and unit conversions recover the simulated group index; it does not validate a real-device model.

5 Fabrication and layout review

To be completed later.

6 Experimental data and data analysis

To be completed later.

7 Conclusion

To be completed later.

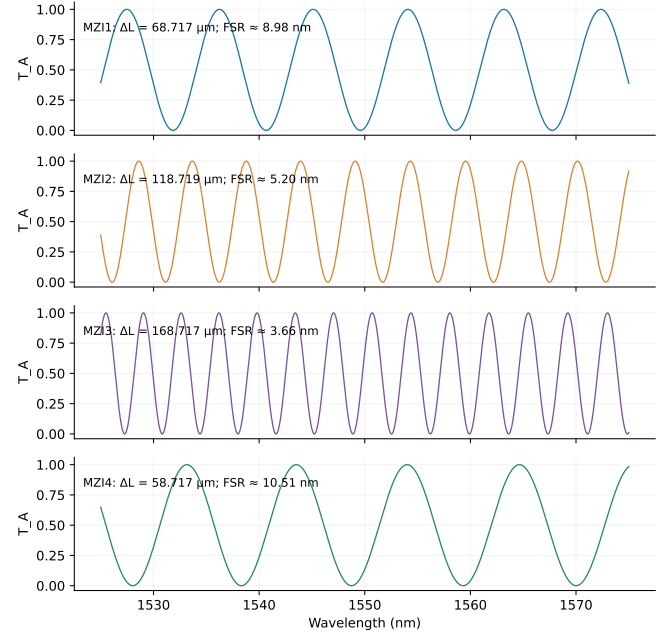


Figure 5: Calculated normalized output-A spectra of all four layout-derived MZIs. These are ideal compact-model simulations with $\alpha = 0$ and $\phi_c = 0$. Coupler/grating S-parameters, reflections, bend losses, and process variations are not included. Actual absolute fringe positions and insertion loss require a calibrated circuit model.

References

- [1] Ansys Optics, “Biosensor circuit,” waveguide dispersion export and unequal-arm MZI model. <https://optics.ansys.com/hc/en-us/articles/360042322634-Biosensor-circuit>, accessed 6 Oct. 2026.
- [2] Z. Lu *et al.*, “Broadband silicon photonic directional coupler using asymmetric-waveguide based phase control,” *Opt. Express*, vol. 23, no. 3, pp. 3795–3806, 2015. doi:10.1364/OE.23.003795.
- [3] Y. Wang *et al.*, “Focusing sub-wavelength grating couplers with low back reflections for rapid prototyping of silicon photonic circuits,” *Opt. Express*, vol. 22, pp. 20652–20662, 2014. doi:10.1364/OE.22.020652. DOI and design attribution are embedded in the grating-coupler cell.